

India's Leap In SUPERCOMPUTING: Innovating For Tomorrow

As India strides towards self-sufficiency in supercomputing, embracing this evolution isn't just an option—it is pivotal for global competitiveness and technological leadership.



PANKAJ PANJWANI, with 25+ years experience in semiconductor design, has delivered 100+ projects in automotive, communication, networking, and other domains. He has held leadership and key positions at MNCs like STMicroelectronics and Genesis.Microchip, and in 2016 founded Keenheads to provide high-value semiconductor design and development capability from India.



Supercomputing architecture is undergoing a transformative shift globally, driven by the need for unprecedented computational power. High performance compute (HPC) machines stand out for their ability to perform complex calculations at speeds far beyond conventional computers. Nations race to harness these capabilities for advancements in weather forecasting, drug discovery, artificial intelligence,

and more. With significant strides in indigenous motherboard design and local manufacturing, coupled with ambitious initiatives like the National Supercomputer Mission, India's journey in supercomputing is the one to watch.

High performance compute (HPC) vs general compute

By definition, an HPC machine must be able to do 10x more computations than